Name:
Roll Number:
Program: **BTech / MTech (2 Year) / MTech (3 Year) / PhD**

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

AI3000: Reinforcement Learning

QUIZ 01

Date: 29 August 2026

Question	1	2	Total
Marks Scored			

1. Consider an MDP with finite horizon $N = 2$, state space $\mathcal{S} = \{0, 1\}$, action space $\mathcal{A} = \{\alpha, \beta\}$, time-homogeneous TPM

$$P(s' \mid s, a) = \begin{cases} 0.8, & s' = 1 - s, a = \alpha, \\ 0.2, & s' = s, a = \alpha, \\ 0.2, & s' = 1 - s, a = \beta, \\ 0.8, & s' = s, a = \beta, \end{cases}$$

and deterministic rewards

$$r(s, a, s') = \begin{cases} \mathrm{XOR}(s, s'), & a = \alpha, \\ \mathrm{AND}(s, s'), & a = \beta\,. \end{cases}$$

Consider a Markov randomized policy $\pi = (\pi_0, \pi_1)$ with transition matrices

$$\begin{aligned} \pi_0(\cdot \mid 0) &= (\overset{\alpha}{1 - \lambda_0}, \overset{\beta}{\lambda_0}), & \pi_0(\cdot \mid 1) &= (\overset{\alpha}{\mu_0}, \overset{\beta}{1 - \mu_0}), \\ \pi_1(\cdot \mid 0) &= (\underset{\alpha}{1 - \lambda_1}, \underset{\beta}{\lambda_1}), & \pi_1(\cdot \mid 1) &= (\underset{\alpha}{\mu_1}, \underset{\beta}{1 - \mu_1}), \end{aligned}$$

(a) **(3 Marks)** Compute the value function $v_2^\pi(s) = \mathbb{E}^\pi[R_1 + R_2 \mid S_0 = s]$ for $s \in \mathcal{S}$.

(b) **(2 Marks)** What are all the optimal policies?

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2. Consider an MDP with finite horizon $N = 2$, state space $\mathcal{S} = \{s\}$, action space $\mathcal{A} = \{a_1, a_2, a_3\}$, and stationary randomized reward R_n where

$$R_n(s, \cdot, s) \quad \begin{array}{ccc} a_1 & a_2 & a_3 \\ \hline 5\ (0.8) & 4\ (0.6) & 10\ (0.3) \\ -5\ (0.2) & 0\ (0.4) & -2\ (0.7) \end{array} \qquad (n = 1, 2),$$

where each entry lists the reward and its probability.

(a) **(1 Mark)** Write down the S-A-S transition probability matrix of the MDP.

(b) **(2 Marks)** Consider the randomized Markov policy $\pi = (\pi_0, \pi_1)$ where

$$\pi_0(\cdot \mid s) = (\overset{a_1}{0.1}, \overset{a_2}{0.3}, \overset{a_3}{0.6}) \qquad \pi_1(\cdot \mid s) = (\overset{a_1}{0.5}, \overset{a_2}{0.2}, \overset{a_3}{0.3})$$

Calculate

$$v_2^\pi(s) = \mathbb{E}^\pi[R_1 + R_2 \mid S_0 = s].$$

(c) **(2 Marks)** Determine an optimal policy π^* and its value

$$v_2^*(s) = \sup_{\pi \in \Pi^{\mathrm{HR}}} v_2^\pi(s).$$